

Replication Output for “Transhumant Pastoralism, Climate Change and Conflict in Africa”

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1. Summary Statistics

Table A3: Descriptive Statistics

	Cell-Year Level Variables, 1989-2018					
	Mean	SD	Count	Min	Median	Max
UCDP: I(Any Conflict), 0/1	0.03	0.18	290,730	0.00	0.00	1.00
UCDP: I(State Conflict), 0/1	0.02	0.15	290,730	0.00	0.00	1.00
UCDP: I(Nonstate Conflict), 0/1	0.02	0.12	290,730	0.00	0.00	1.00
ACLED: I(Any Conflict), 0/1	0.08	0.27	213,202	0.00	0.00	1.00
ACLED: I(State Conflict), 0/1	0.05	0.22	213,202	0.00	0.00	1.00
ACLED: I(Nonstate Conflict), 0/1	0.08	0.27	213,202	0.00	0.00	1.00
Precipitation, cm/month	5.65	5.14	290,730	0.00	4.38	49.28
Phytomass, kg/ha/day	30.69	30.35	193,820	0.01	23.44	141.11
Temperature, °C	24.50	3.95	251,922	7.51	24.75	39.53
Nearest Neighbor Precipitation, cm/month	5.89	5.06	282,690	0.00	4.83	34.96
Nearest Neighbor Phytomass, kg/ha/day	31.90	29.77	188,460	0.18	25.77	130.71
Nearest Neighbor Temperature, °C	24.44	3.79	244,998	12.20	24.65	37.12
Nighttime Lights, 0-1	0.04	0.03	203,511	0.00	0.03	0.96

	Cell Level Variables					
	Mean	SD	Count	Min	Median	Max
Nearest Neighbor Transhumant Pastoralism (Narrow Definition), 0-1	0.19	0.30	8,487	0.00	0.00	0.92
Nearest Neighbor Transhumant Pastoralism (Broad Definition), 0-1	0.21	0.30	8,487	0.00	0.00	0.92
B-S: Land Suitability for Transhumant Pastoralism, 0-1	0.32	0.20	9,421	0.00	0.29	0.90
B-S: Land Suitability for Agriculture, 0-1	0.24	0.20	9,421	0.00	0.22	0.88
ln(Population) in 1990	9.31	2.11	9,659	-0.69	9.61	16.01

Ethnic-Group-Year Level Variables, 1989-2018						
	Mean	SD	Count	Min	Median	Max
Precipitation, cm/month	8.54	5.20	23,400	0.00	8.27	34.96
Phytomass, kg/ha/day	44.31	28.53	15,600	0.18	43.59	130.71
Temperature, °C	24.78	3.47	20,280	12.20	25.28	37.12
EPR: Political Power, 0-5	2.13	1.16	12,500	0.00	2.00	5.00

	Ethnic Group Level Variables					
	Mean	SD	Count	Min	Median	Max
Transhumant Pastoralism (Narrow Definition), 0-1	0.08	0.22	712	0.00	0.00	0.92
Transhumant Pastoralism (Broad Definition), 0-1	0.09	0.23	712	0.00	0.00	0.92
EA: Agriculture, 0-1	0.55	0.18	745	0.03	0.61	0.92
EA: Jurisdictional Hierarchy, 0-4	1.29	0.97	687	0.00	1.00	4.00
EA: Belief in High Gods, 0/1	0.45	0.50	488	0.00	0.00	1.00
Share Muslim, 0-1	0.29	0.38	689	0.00	0.05	1.00
Share Christian, 0-1	0.46	0.35	689	0.00	0.46	1.00
Segmentary Lineage, 0-1	0.50	0.25	722	0.02	0.48	0.98

Table A4: Balance Table

Variable	(1) THP > 0	(2) THP = 0	(3) Difference
UCDP: I(Any Conflict), 0/1	0.024 (0.152)	0.041 (0.198)	-0.017*** (0.002)
UCDP: I(State Conflict), 0/1	0.017 (0.130)	0.029 (0.168)	-0.012*** (0.002)
UCDP: I(Nonstate Conflict), 0/1	0.009 (0.095)	0.020 (0.140)	-0.011*** (0.001)
ACLED: I(Any Conflict), 0/1	0.051 (0.221)	0.098 (0.297)	-0.047*** (0.003)
ACLED: I(State Conflict), 0/1	0.034 (0.180)	0.063 (0.243)	-0.030*** (0.002)
ACLED: I(Nonstate Conflict), 0/1	0.051 (0.220)	0.098 (0.297)	-0.047*** (0.003)
Precipitation, cm/month	2.066 (2.715)	8.513 (4.857)	-6.447*** (0.078)
Phytomass, kg/ha/day	9.214 (17.333)	47.835 (27.446)	-38.621*** (0.475)
Temperature, °C	25.323 (4.115)	23.859 (3.688)	1.465*** (0.083)
Nearest Neighbor Precipitation, cm/month	2.400 (2.843)	8.531 (4.768)	-6.131*** (0.080)
Nearest Neighbor Phytomass, kg/ha/day	11.216 (17.965)	47.620 (27.075)	-36.404*** (0.484)
Nearest Neighbor Temperature, °C	25.213 (3.939)	23.879 (3.542)	1.334*** (0.081)
Nighttime Lights, 0-1	0.037 (0.021)	0.042 (0.043)	-0.006*** (0.001)
Observations	115,650	148,740	290,730

Variable	(1) THP > 0	(2) THP = 0	(3) Difference
Nearest Neighbor Transhumant Pastoralism (Narrow Definition), 0-1	0.357 (0.333)	0.070 (0.204)	0.287*** (0.006)
Nearest Neighbor Transhumant Pastoralism (Broad Definition), 0-1	0.378 (0.323)	0.085 (0.214)	0.294*** (0.006)
B-S: Land Suitability for Transhumant Pastoralism, 0-1	0.390 (0.196)	0.266 (0.186)	0.124*** (0.004)
B-S: Land Suitability for Agriculture, 0-1	0.099 (0.132)	0.354 (0.182)	-0.255*** (0.004)
ln(Population) in 1990	8.093 (1.977)	10.230 (1.729)	-2.137*** (0.040)
Observations	3,855	4,958	9,691

Variable	(1) THP > 0	(2) THP = 0	(3) Difference
Precipitation, cm/month	3.840 (3.342)	9.745 (4.885)	-5.905*** (0.349)
Phytomass, kg/ha/day	19.923 (23.412)	50.563 (26.176)	-30.640*** (2.339)
Temperature, °C	25.171 (4.014)	24.756 (3.330)	0.415 (0.377)
EPR: Political Power, 0-5	1.894 (1.237)	2.169 (1.093)	-0.274** (0.135)
Observations	3,750	17,610	23,400

Variable	(1) THP > 0	(2) THP = 0	(3) Difference
EA: Agriculture, 0-1	0.338 (0.208)	0.593 (0.133)	-0.255*** (0.015)
EA: Jurisdictional Hierarchy, 0-4	1.555 (0.852)	1.240 (0.980)	0.315*** (0.100)
EA: Belief in High Gods, 0/1	0.779 (0.417)	0.355 (0.479)	0.424*** (0.050)
Share Muslim, 0-1	0.565 (0.478)	0.246 (0.337)	0.319*** (0.039)
Share Christian, 0-1	0.278 (0.361)	0.484 (0.339)	-0.205*** (0.037)
Segmentary Lineage, 0-1	0.476 (0.191)	0.509 (0.257)	-0.033 (0.025)
Observations	125	587	780

Table A16: Descriptive Statistics for Country-Year Level Variables

	Country-Year Level Variables					
	Mean	SD	Count	Min	Median	Max
Total Agriculture Aid	3.87	8.56	1,421	0.00	0.97	97.40
Total Non-Agriculture Aid	52.36	121.44	1,421	0.00	11.78	1176.00
Irrigation Projects	0.41	0.81	1,421	0.00	0.11	7.67
Forestry Projects	0.88	1.75	1,421	0.00	0.25	17.00
Conservation Projects	0.50	1.14	1,421	0.00	0.10	12.33
Land Projects	0.47	1.09	1,421	0.00	0.11	13.00
Share Protected Area in Country	0.14	0.10	1,764	0.00	0.12	0.54
THP Power Share	0.10	0.16	1,053	0.00	0.00	0.61

2. Main Tables

Table 1: Cross-Sectional Evidence of Conflict Spillover from Nearest Neighboring THP Territory: Cell Level

	UCDP			ACLED		
	(1) I(Any)	(2) I(State)	(3) I(Nonstate)	(4) I(Any)	(5) I(State)	(6) I(Nonstate)
Neighbor Transhumant Pastoral [γ_1]	0.0273*** (0.0054)	0.0253*** (0.0049)	0.0057** (0.0025)	0.0702*** (0.0095)	0.0534*** (0.0077)	0.0698*** (0.0095)
Transhumant Pastoral [γ_2]	0.0081 (0.0057)	0.0063 (0.0046)	0.0017 (0.0028)	0.0208** (0.0097)	0.0134* (0.0077)	0.0200** (0.0097)
Dep. Var. Mean	0.035	0.025	0.016	0.085	0.055	0.085
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	336	336	336
Cell Clusters	7,690	7,690	7,690	7,690	7,690	7,690
Observations	230,700	230,700	230,700	184,560	184,560	184,560

	UCDP			ACLED		
	(1) I(Any)	(2) I(State)	(3) I(Nonstate)	(4) I(Any)	(5) I(State)	(6) I(Nonstate)
Neighbor Transhumant Pastoral [γ_1]	0.0301*** (0.0053)	0.0288*** (0.0049)	0.0050** (0.0023)	0.0671*** (0.0089)	0.0534*** (0.0074)	0.0667*** (0.0089)
Transhumant Pastoral [γ_2]	0.0076 (0.0054)	0.0058 (0.0044)	0.0012 (0.0026)	0.0188** (0.0093)	0.0126* (0.0074)	0.0181* (0.0093)
Dep. Var. Mean	0.035	0.025	0.016	0.085	0.055	0.085
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	336	336	336
Cell Clusters	7,690	7,690	7,690	7,690	7,690	7,690
Observations	230,700	230,700	230,700	184,560	184,560	184,560

Table 2: Effect of Rain Shock in Nearest Neighboring THP Territory on Conflict in a Cell: Narrow Definition of Transhumance

	Indicator for the presence of conflict					
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) UCDP I(Nonstate)	(4) ACLED I(Any)	(5) ACLED I(State)	(6) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>						
Rain [γ_0^s]	-0.0005 (0.0006)	0.0001 (0.0006)	-0.0005 (0.0005)	-0.0007 (0.0011)	0.0004 (0.0009)	-0.0008 (0.0011)
Rain $\times$ Transhumant Pastoral [γ_1^s]	-0.0110*** (0.0033)	-0.0121*** (0.0031)	-0.0012 (0.0021)	-0.0096** (0.0038)	-0.0092*** (0.0035)	-0.0096** (0.0038)
<i>Own Ethnic Group</i>						
Rain [γ_2^s]	0.0001 (0.0010)	0.0014 (0.0009)	-0.0002 (0.0007)	0.0007 (0.0013)	0.0014 (0.0010)	0.0005 (0.0013)
Rain $\times$ Transhumant Pastoral [γ_3^s]	-0.0014 (0.0047)	-0.0046 (0.0048)	0.0017 (0.0038)	-0.0011 (0.0065)	-0.0079 (0.0062)	0.0005 (0.0065)
<i>Own Cell</i>						
Rain [γ_4^s]	-0.0002 (0.0007)	-0.0005 (0.0006)	-0.0001 (0.0005)	-0.0004 (0.0010)	-0.0007 (0.0009)	-0.0002 (0.0010)
Rain $\times$ Transhumant Pastoral [γ_5^s]	0.0041 (0.0035)	0.0056* (0.0032)	-0.0008 (0.0024)	0.0046 (0.0051)	0.0052 (0.0039)	0.0032 (0.0051)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>						
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:						
Rain	-1.88	0.57	-3.51	-0.95	0.83	-1.13
p-value	[0.40]	[0.83]	[0.36]	[0.53]	[0.67]	[0.46]
Rain $\times$ Transhumant Pastoral	-37.51	-57.26	-8.68	-13.60	-20.12	-13.64
p-value	[0.00]	[0.00]	[0.58]	[0.01]	[0.01]	[0.01]
Rain + Rain $\times$ Transhumant Pastoral	-39.39	-56.68	-12.19	-14.55	-19.29	-14.76
p-value	[0.00]	[0.00]	[0.43]	[0.01]	[0.01]	[0.00]
Dep. Var. Mean	0.035	0.025	0.016	0.085	0.055	0.084
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	322	322	322
Cell Clusters	7,722	7,722	7,722	7,722	7,722	7,722
Observations	231,660	231,660	231,660	177,606	177,606	177,606

Table 3: Robustness to Controlling for the Components of Transhumant Pastoralism

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0014 (0.0011)	0.0005 (0.0008)	-0.0020 (0.0015)	-0.0021 (0.0015)
Rain $\times$ Pastoral	0.0043 (0.0044)	-0.0022 (0.0035)	0.0069 (0.0061)	0.0067 (0.0061)
Rain $\times$ Transhumant	0.0040 (0.0025)	0.0020 (0.0018)	0.0029 (0.0039)	0.0031 (0.0039)
Rain $\times$ Transhumant Pastoral	-0.0191*** (0.0069)	-0.0128** (0.0057)	-0.0186** (0.0088)	-0.0187** (0.0088)
Dep. Var. Mean	0.0351	0.0253	0.0845	0.0842
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	7,722	7,722	7,722	7,722
Observations	231,660	231,660	177,606	177,606

Table 4: Effect of Rain in Nearest Neighboring THP Territory on Conflict in Agricultural and Non-Agricultural Cells

	Conflict in Agricultural Cells				Conflict in Non-Agricultural Cells			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)	(5) UCDP I(Any)	(6) UCDP I(State)	(7) ACLED I(Any)	(8) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>								
Rain [γ_0^s]	-0.0006 (0.0007)	0.0002 (0.0006)	-0.0002 (0.0011)	-0.0004 (0.0011)	0.0000 (0.0026)	-0.0001 (0.0024)	-0.0105*** (0.0036)	-0.0103*** (0.0036)
Rain $\times$ Transhumant Pastoral [γ_1^s]	-0.0119** (0.0047)	-0.0121*** (0.0039)	-0.0172*** (0.0056)	-0.0180*** (0.0057)	-0.0053 (0.0056)	-0.0062 (0.0051)	0.0052 (0.0064)	0.0056 (0.0064)
Dep. Var. Mean	0.039	0.028	0.097	0.096	0.025	0.019	0.055	0.055
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	390	390	299	299	390	390	299	299
Cell Clusters	5,482	5,482	5,482	5,482	2,240	2,240	2,240	2,240
Observations	164,460	164,460	126,086	126,086	67,200	67,200	51,520	51,520

Table 5: Estimates Using Phytomass and Temperature Rather than Rainfall

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Phytomass	0.0001 (0.0005)	0.0001 (0.0004)	0.0003 (0.0006)	0.0004 (0.0006)
Phytomass $\times$ Transhumant Pastoral	-0.0043** (0.0018)	-0.0041** (0.0016)	-0.0085*** (0.0018)	-0.0086*** (0.0018)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Shock as % of Dep. Var. Mean:				
Shock $\times$ Transhumant Pastoral p-value	-37.95 [0.02]	-50.73 [0.01]	-32.09 [0.00]	-32.70 [0.00]
Dep. Var. Mean	0.037	0.027	0.087	0.087
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	280	280	294	294
Cell Clusters	7,722	7,722	7,722	7,722
Observations	154,440	154,440	162,162	162,162
	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Temperature	0.0019 (0.0016)	0.0028** (0.0013)	0.0027 (0.0027)	0.0026 (0.0027)
Temperature $\times$ Transhumant Pastoral	0.0022 (0.0037)	0.0047 (0.0035)	0.0030 (0.0045)	0.0029 (0.0045)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Shock as % of Dep. Var. Mean:				
Shock $\times$ Transhumant Pastoral p-value	5.55 [0.54]	15.89 [0.18]	3.48 [0.52]	3.42 [0.52]
Dep. Var. Mean	0.032	0.024	0.068	0.068
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	364	364	252	252
Cell Clusters	7,722	7,722	7,722	7,722
Observations	200,728	200,728	138,968	138,968

Table 6: Effects of Neighbor's Rainfall on Conflict during the Wet and Dry Seasons

	Growing Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Annual Rain	0.0008 (0.0023)	0.0039 (0.0048)	-0.0022 (0.0032)	-0.0017 (0.0113)
Annual Rain $\times$ Transhumant Pastoral	-0.0346*** (0.0128)	-0.1294** (0.0613)	-0.0152 (0.0116)	-0.0669 (0.0444)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain $\times$ Transhumant Pastoral	-46.32	-94.31	-18.60	-44.40
p-value	[0.01]	[0.04]	[0.19]	[0.13]
Dep. Var. Mean	0.090	0.165	0.098	0.181
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	420
Cell Clusters	4,632	4,632	4,632	4,632
Observations	138,960	138,960	138,960	138,960
	Growing Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Wet Season Rain	0.0013 (0.0016)	0.0037 (0.0041)		
Wet Season Rain $\times$ Transhumant Pastoral	-0.0184* (0.0104)	-0.0834* (0.0486)		
Dry Season Rain			-0.0015 (0.0022)	0.0008 (0.0073)
Dry Season Rain $\times$ Transhumant Pastoral			-0.0043 (0.0103)	-0.0103 (0.0214)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain $\times$ Transhumant Pastoral	-41.73	-102.80	-6.75	-8.69
p-value	[0.08]	[0.09]	[0.68]	[0.63]
Dep. Var. Mean	0.090	0.165	0.098	0.181
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	420
Cell Clusters	4,632	4,632	4,632	4,632
Observations	138,960	138,960	138,960	138,960

Table 7: Jihadist Violence

	Indicator for presence of conflict			
	(1) I(Jihadist)	(2) I(Non-Jihadist)	(3) I(Jihadist)	(4) I(Non-Jihadist)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0000 (0.0003)	-0.0006 (0.0006)	0.0006 (0.0005)	0.0002 (0.0020)
Rain × Transhumant Pastoral	-0.0051** (0.0022)	-0.0063** (0.0026)	-0.0056** (0.0025)	-0.0056* (0.0030)
Rain × Share Muslim			-0.0020 (0.0015)	-0.0016 (0.0025)
Rain × Share Christian			-0.0003 (0.0006)	-0.0007 (0.0028)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain × Transhumant Pastoral p-value	-82.03 [0.02]	-27.05 [0.01]	-82.42 [0.03]	-21.41 [0.06]
Dep. Var. Mean	0.007	0.028	0.008	0.032
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	420
Cell Clusters	7,722	7,722	6,507	6,507
Observations	231,660	231,660	195,210	195,210

Table 8: Heterogeneity by the Presence of International Aid Projects

	Indicator for presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral	-0.0129*** (0.0038)	-0.0122*** (0.0035)	-0.0038 (0.0039)	-0.0036 (0.0039)
Rain × Transhumant Pastoral × Total Agriculture Aid	-0.0059 (0.0064)	-0.0068 (0.0055)	-0.0113 (0.0074)	-0.0117 (0.0075)
Rain × Transhumant Pastoral × Total Non-Agriculture Aid	0.0004 (0.0004)	0.0004 (0.0004)	0.0005 (0.0005)	0.0005 (0.0005)
Dep. Var. Mean	0.032	0.024	0.068	0.068
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	364	364	252	252
Cell Clusters	7,722	7,722	7,722	7,722
Observations	200,772	200,772	138,996	138,996

Table 9: Heterogeneity by the Share of Conservation Lands in a Country

	Indicator for presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral	-0.0082 (0.0050)	-0.0073 (0.0047)	-0.0005 (0.0055)	-0.0003 (0.0055)
Rain × Transhumant Pastoral × Share Protected Area in Country	-0.0248 (0.0259)	-0.0390 (0.0258)	-0.0626** (0.0260)	-0.0638** (0.0262)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain × Transhumant Pastoral when Protected Area at 10th pctl	-28.6	-35.6	-1.1	-0.9
p-value	[0.09]	[0.10]	[0.88]	[0.91]
Rain × Transhumant Pastoral when Protected Area at 90th pctl	-52.2	-87.0	-25.9	-26.2
p-value	[0.00]	[0.00]	[0.00]	[0.00]
Dep. Var. Mean	0.035	0.025	0.085	0.084
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	7,718	7,718	7,718	7,718
Observations	231,540	231,540	177,514	177,514

Table 10: Heterogeneity by Share of Political Power Held by Transhumant Pastoral Groups

	Indicator for presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral	-0.0158** (0.0062)	-0.0151*** (0.0054)	-0.0513*** (0.0091)	-0.0513*** (0.0091)
Rain × Transhumant Pastoral × THP Power Share	0.0458** (0.0231)	0.0367* (0.0211)	0.1834*** (0.0392)	0.1824*** (0.0393)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain × Transhumant Pastoral when THP Power at 10th pctl	-58.1	-74.0	-81.7	-82.1
p-value	[0.01]	[0.01]	[0.00]	[0.00]
Rain × Transhumant Pastoral when THP Power at 90th pctl	-7.2	-19.4	6.8	6.3
p-value	[0.64]	[0.32]	[0.52]	[0.56]
Dep. Var. Mean	0.033	0.024	0.075	0.075
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Years	406	406	308	308
Cells	7,018	7,018	7,015	7,015
Observations	195,975	195,975	149,290	149,290

3. Rest of Appendix Tables

Table A2: The Determinants of Phytomass Growth

	Phytomass		
	(1)	(2)	(3)
Rain	0.4151*** (0.0357)		0.4092*** (0.0350)
Temp		-0.2223*** (0.0400)	-0.2018*** (0.0383)
Share of RSS explained by weather variable(s) (in %)	3.63	0.61	4.13
F statistic	135.55	30.84	75.07
Effect of 1 Std. Dev. Shock as % of Dep. Var. Mean:			
Rain p-value	1.63 [0.00]		1.61 [0.00]
Temp p-value		-0.58 [0.00]	-0.53 [0.00]
Dep. Var. Mean	30.57	30.57	30.57
Cell FE	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes
Climate-Zone-Years	224	224	224
Cells	9,691	9,691	9,691
Observations	155,032	155,032	155,032

Table A5: Effect of Rain Shock in Nearest Neighboring THP Territory on Conflict in a Cell: Broad Definition of Transhumance

	Indicator for the presence of conflict					
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) UCDP I(Nonstate)	(4) ACLED I(Any)	(5) ACLED I(State)	(6) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>						
Rain [γ_0^s]	-0.0005 (0.0006)	0.0002 (0.0006)	-0.0006 (0.0005)	-0.0005 (0.0011)	0.0005 (0.0009)	-0.0007 (0.0011)
Rain $\times$ Transhumant Pastoral [γ_1^s]	-0.0082*** (0.0031)	-0.0105*** (0.0028)	0.0008 (0.0019)	-0.0093** (0.0037)	-0.0081** (0.0036)	-0.0094** (0.0037)
<i>Own Ethnic Group</i>						
Rain [γ_2^s]	0.0003 (0.0010)	0.0016* (0.0009)	-0.0001 (0.0007)	0.0008 (0.0014)	0.0015 (0.0011)	0.0006 (0.0014)
Rain $\times$ Transhumant Pastoral [γ_3^s]	-0.0050 (0.0042)	-0.0065 (0.0042)	-0.0009 (0.0035)	-0.0028 (0.0062)	-0.0080 (0.0057)	-0.0013 (0.0062)
<i>Own Cell</i>						
Rain [γ_4^s]	-0.0004 (0.0007)	-0.0006 (0.0006)	-0.0002 (0.0005)	-0.0005 (0.0010)	-0.0008 (0.0009)	-0.0003 (0.0010)
Rain $\times$ Transhumant Pastoral [γ_5^s]	0.0049 (0.0033)	0.0062** (0.0030)	0.0001 (0.0024)	0.0054 (0.0048)	0.0056 (0.0039)	0.0041 (0.0048)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>						
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:						
Rain	-1.82	1.03	-4.13	-0.77	1.01	-0.95
p-value	[0.41]	[0.70]	[0.29]	[0.61]	[0.61]	[0.54]
Rain $\times$ Transhumant Pastoral	-27.95	-49.97	6.15	-13.27	-17.70	-13.36
p-value	[0.01]	[0.00]	[0.67]	[0.01]	[0.02]	[0.01]
Rain + Rain $\times$ Transhumant Pastoral	-29.77	-48.95	2.01	-14.04	-16.69	-14.31
p-value	[0.00]	[0.00]	[0.89]	[0.01]	[0.03]	[0.01]
Dep. Var. Mean	0.035	0.025	0.016	0.085	0.055	0.084
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	420	322	322	322
Cell Clusters	7,722	7,722	7,722	7,722	7,722	7,722
Observations	231,660	231,660	231,660	177,606	177,606	177,606

Table A6: Robustness to Additional Controls for Ethnicity-Level Characteristics

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0025 (0.0015)	-0.0010 (0.0013)	-0.0025 (0.0023)	-0.0026 (0.0023)
Rain $\times$ Transhumant Pastoral	-0.0117*** (0.0036)	-0.0123*** (0.0031)	-0.0094** (0.0038)	-0.0097** (0.0038)
Rain $\times$ Jurisdictional Hierarchy	0.0004 (0.0006)	-0.0000 (0.0005)	-0.0002 (0.0008)	-0.0002 (0.0008)
Rain $\times$ Segmentary Lineage	0.0028 (0.0019)	0.0022 (0.0015)	0.0030 (0.0029)	0.0029 (0.0030)
Rain $\times$ High Gods: Active, Not Supportive	0.0013 (0.0023)	0.0015 (0.0016)	0.0030 (0.0036)	0.0031 (0.0036)
Rain $\times$ High Gods: Active, Supportive	0.0014 (0.0013)	0.0017* (0.0011)	-0.0010 (0.0022)	-0.0009 (0.0022)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain	-8.52	-4.86	-3.41	-3.61
p-value	[0.10]	[0.43]	[0.28]	[0.26]
Rain $\times$ Transhumant Pastoral	-39.42	-59.20	-13.00	-13.44
p-value	[0.00]	[0.00]	[0.01]	[0.01]
Rain + Rain $\times$ Transhumant Pastoral	-47.93	-64.06	-16.42	-17.05
p-value	[0.00]	[0.00]	[0.01]	[0.00]
Dep. Var. Mean	0.0357	0.0249	0.0869	0.0865
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	6,603	6,603	6,603	6,603
Observations	198,090	198,090	151,869	151,869

Table A7: Robustness to Additional Controls for Ethnicity-Level Average Rainfall

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0000 (0.0014)	-0.0004 (0.0013)	-0.0004 (0.0021)	-0.0004 (0.0021)
Rain $\times$ Transhumant Pastoral	-0.0111*** (0.0034)	-0.0116*** (0.0031)	-0.0100** (0.0040)	-0.0100** (0.0039)
Rain $\times$ Average Rain (0-1)	-0.0012 (0.0028)	0.0012 (0.0028)	-0.0008 (0.0042)	-0.0009 (0.0042)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain	-0.10	-1.80	-0.52	-0.62
p-value	[0.98]	[0.78]	[0.86]	[0.84]
Rain $\times$ Transhumant Pastoral	-37.91	-55.18	-14.26	-14.30
p-value	[0.00]	[0.00]	[0.01]	[0.01]
Rain + Rain $\times$ Transhumant Pastoral	-38.01	-56.97	-14.78	-14.92
p-value	[0.00]	[0.00]	[0.01]	[0.01]
Dep. Var. Mean	0.0351	0.0253	0.0845	0.0842
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Years	420	420	322	322
Cells	7,722	7,722	7,722	7,722
Observations	231,660	231,660	177,606	177,606

Table A8: Robustness to Additional Controls for Time-Varying Characteristics

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0006 (0.0006)	0.0001 (0.0006)	-0.0007 (0.0011)	-0.0009 (0.0011)
Rain $\times$ Transhumant Pastoral	-0.0114*** (0.0034)	-0.0126*** (0.0031)	-0.0095*** (0.0036)	-0.0094*** (0.0036)
Year $\times$ Transhumant Pastoral	-0.0005 (0.0006)	-0.0003 (0.0006)	-0.0064*** (0.0018)	-0.0064*** (0.0017)
Price Index: Energy $\times$ Transhumant Pastoral	0.0006*** (0.0002)	0.0004** (0.0002)	0.0005** (0.0002)	0.0005** (0.0002)
Price Index: Metals and Minerals $\times$ Transhumant Pastoral	0.0001 (0.0002)	0.0003 (0.0002)	-0.0004 (0.0003)	-0.0004 (0.0003)
Price Index: Precious Metals $\times$ Transhumant Pastoral	-0.0004 (0.0003)	-0.0005* (0.0002)	0.0005 (0.0005)	0.0005 (0.0005)
Price Index: Agriculture $\times$ Transhumant Pastoral	-0.0001 (0.0005)	0.0001 (0.0004)	0.0006 (0.0007)	0.0006 (0.0007)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain	-1.90	0.61	-1.04	-1.22
p-value	[0.39]	[0.82]	[0.49]	[0.42]
Rain $\times$ Transhumant Pastoral	-39.14	-59.85	-13.46	-13.45
p-value	[0.00]	[0.00]	[0.01]	[0.01]
Rain + Rain $\times$ Transhumant Pastoral	-41.04	-59.25	-14.50	-14.67
p-value	[0.00]	[0.00]	[0.00]	[0.00]
Dep. Var. Mean	0.0351	0.0253	0.0845	0.0842
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	7,722	7,722	7,722	7,722
Observations	231,660	231,660	177,606	177,606

Table A9: Robustness to Various Inference Procedures

	Panel A: Clustering by Country			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0005 (0.0006)	0.0001 (0.0006)	-0.0007 (0.0011)	-0.0008 (0.0011)
Rain $\times$ Transhumant Pastoral	-0.0110** (0.0044)	-0.0121*** (0.0038)	-0.0096*** (0.0022)	-0.0096*** (0.0023)
Country Clusters	49	49	49	49
	Panel B: Clustering by Country and Climate-Zone			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0005 (0.0006)	0.0001 (0.0005)	-0.0007 (0.0010)	-0.0008 (0.0010)
Rain $\times$ Transhumant Pastoral	-0.0110*** (0.0033)	-0.0121*** (0.0028)	-0.0096*** (0.0014)	-0.0096*** (0.0015)
Country Clusters	49	49	49	49
Climate-Zone Clusters	14	14	14	14
	Panel C: Spatial HAC 1000KM			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0005 (0.0007)	0.0001 (0.0006)	-0.0007 (0.0010)	-0.0008 (0.0010)
Rain $\times$ Transhumant Pastoral	-0.0110*** (0.0040)	-0.0121*** (0.0035)	-0.0096** (0.0043)	-0.0096** (0.0043)
Observations	231,660	231,660	177,606	177,606

Table A10: Phytomass Suitability Index (Predicted by Rain)

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Phytomass Index [$f(rain)$]	-0.0001 (0.0001)	0.0000 (0.0001)	-0.0001 (0.0002)	-0.0002 (0.0002)
Phytomass Index [$f(rain)$] $\times$ Transhumant Pastoral	-0.0021*** (0.0006)	-0.0023*** (0.0006)	-0.0018** (0.0007)	-0.0018** (0.0007)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Suitability Index Shock as % of Dep. Var. Mean:				
Phytomass Suitability Index p-value	-1.88 [0.40]	0.57 [0.83]	-0.95 [0.53]	-1.13 [0.46]
Phytomass Suitability Index $\times$ Transhumant Pastoral p-value	-37.51 [0.00]	-57.26 [0.00]	-13.60 [0.01]	-13.64 [0.01]
Phytomass Suitability Index + Phytomass Suitability Index $\times$ Transhumant Pastoral p-value	-39.39 [0.00]	-56.68 [0.00]	-14.55 [0.01]	-14.76 [0.00]
Dep. Var. Mean	0.035	0.025	0.085	0.084
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	7,722	7,722	7,722	7,722
Observations	231,660	231,660	177,606	177,606

Table A11: Phytomass Suitability Index (Predicted by Rain and Rain Squared)

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Phytomass Index [$f(rain, rain^2)$]	-0.0001 (0.0002)	0.0000 (0.0001)	-0.0002 (0.0002)	-0.0002 (0.0002)
Phytomass Index [$f(rain, rain^2)$] $\times$ Transhumant Past.	-0.0018*** (0.0005)	-0.0020*** (0.0005)	-0.0014** (0.0006)	-0.0014** (0.0006)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Suitability Index Shock as % of Dep. Var. Mean:				
Phytomass Suitability Index p-value	-0.94 [0.70]	0.25 [0.93]	-1.23 [0.42]	-1.33 [0.39]
Phytomass Suitability Index $\times$ Transhumant Pastoral p-value	-28.07 [0.00]	-43.06 [0.00]	-9.20 [0.02]	-9.17 [0.02]
Phytomass Suitability Index + Phytomass Suitability Index $\times$ Transhumant Pastoral p-value	-29.01 [0.00]	-42.80 [0.00]	-10.44 [0.01]	-10.50 [0.01]
Dep. Var. Mean	0.035	0.025	0.085	0.084
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	7,722	7,722	7,722	7,722
Observations	231,660	231,660	177,606	177,606

Table A12: Allowing for Rainfall and Temperature in the Same Specification

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	-0.0004 (0.0007)	0.0002 (0.0006)	-0.0002 (0.0011)	-0.0004 (0.0012)
Rain $\times$ Transhumant Pastoral	-0.0116*** (0.0035)	-0.0125*** (0.0033)	-0.0071** (0.0036)	-0.0075** (0.0035)
Temperature	0.0020 (0.0016)	0.0029** (0.0013)	0.0028 (0.0027)	0.0026 (0.0027)
Temperature $\times$ Transhumant Pastoral	0.0017 (0.0037)	0.0040 (0.0035)	0.0026 (0.0045)	0.0026 (0.0044)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain	-1.66	1.18	-0.41	-0.72
p-value	[0.50]	[0.68]	[0.84]	[0.72]
Rain $\times$ Transhumant Pastoral	-42.99	-63.49	-12.53	-13.18
p-value	[0.00]	[0.00]	[0.05]	[0.03]
Rain + Rain $\times$ Transhumant Pastoral	-44.65	-62.31	-12.94	-13.90
p-value	[0.00]	[0.00]	[0.03]	[0.02]
Effect of 1 Std. Dev. Temp Shock as % of Dep. Var. Mean:				
Temp	7.24	14.83	4.87	4.65
p-value	[0.23]	[0.03]	[0.30]	[0.33]
Temp $\times$ Transhumant Pastoral	6.23	20.28	4.62	4.51
p-value	[0.65]	[0.25]	[0.56]	[0.56]
Temp + Temp $\times$ Transhumant Pastoral	13.47	35.11	9.49	9.17
p-value	[0.32]	[0.05]	[0.17]	[0.17]
Dep. Var. Mean	0.032	0.024	0.068	0.068
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	364	364	252	252
Cell Clusters	7,722	7,722	7,722	7,722
Observations	200,728	200,728	138,968	138,968

Table A13: Effects of Neighbor's Phytomass on Conflict during the Wet and Dry Seasons

	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Annual Phytomass	0.0009 (0.0015)	0.0037 (0.0035)	0.0009 (0.0015)	0.0034 (0.0033)
Annual Phytomass $\times$ Transhumant Pastoral	-0.0089** (0.0041)	-0.0385* (0.0219)	-0.0009 (0.0044)	-0.0155 (0.0175)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-30.71 [0.03]	-70.21 [0.08]	-2.87 [0.84]	-25.72 [0.37]
Dep. Var. Mean	0.096	0.182	0.106	0.200
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	280	280	280	280
Cell Clusters	4,632	4,632	4,632	4,632
Observations	92,640	92,640	92,640	92,640
	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Wet Season Phytomass	0.0006 (0.0012)	-0.0000 (0.0036)		
Wet Season Phytomass $\times$ Transhumant Pastoral	-0.0064** (0.0032)	-0.0218* (0.0128)		
Dry Season Phytomass			0.0006 (0.0012)	-0.0006 (0.0056)
Dry Season Phytomass $\times$ Transhumant Pastoral			-0.0030 (0.0042)	-0.0301 (0.0223)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-32.56 [0.05]	-57.92 [0.09]	-12.40 [0.48]	-65.71 [0.18]
Dep. Var. Mean	0.096	0.182	0.106	0.200
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	280	280	280	280
Cell Clusters	4,632	4,632	4,632	4,632
Observations	92,640	92,640	92,640	92,640

Table A14 Panel A: Summary of Seasonal Regressions, Agricultural Cells Only, Rain

	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Rain	0.0005 (0.0024)	0.0042 (0.0050)	-0.0025 (0.0033)	-0.0034 (0.0124)
Rain × Transhumant Pastoral	-0.0361** (0.0173)	-0.0918* (0.0472)	-0.0029 (0.0162)	-0.0262 (0.0499)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain × Transhumant Pastoral p-value	-49.64 [0.04]	-72.94 [0.05]	-3.56 [0.86]	-18.14 [0.60]
Dep. Var. Mean	0.087	0.151	0.098	0.174
Cell FE	Yes	Yes	Yes	Yes
Country × Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	390	390	390	390
Cell Clusters	3,897	3,897	3,897	3,897
Observations	116,910	116,910	116,910	116,910
	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Wet Season Rain	0.0011 (0.0015)	0.0062 (0.0038)		
Wet Season Rain × Transhumant Pastoral	-0.0229* (0.0134)	-0.0603* (0.0359)		
Dry Season Rain			-0.0021 (0.0023)	-0.0048 (0.0061)
Dry Season Rain × Transhumant Pastoral			-0.0067 (0.0169)	-0.0199 (0.0272)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass × Transhumant Pastoral p-value	-53.30 [0.09]	-80.99 [0.09]	-10.42 [0.69]	-17.47 [0.47]
Dep. Var. Mean	0.087	0.151	0.098	0.174
Cell FE	Yes	Yes	Yes	Yes
Country × Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	390	390	390	390
Cell Clusters	3,897	3,897	3,897	3,897
Observations	116,910	116,910	116,910	116,910

Table A14 Panel B: Summary of Seasonal Regressions, Agricultural Cells Only, Phytomass

	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Annual Phytomass	0.0001 (0.0017)	0.0041 (0.0037)	0.0006 (0.0018)	0.0033 (0.0039)
Annual Phytomass $\times$ Transhumant Pastoral	-0.0095* (0.0053)	-0.0196*** (0.0070)	0.0041 (0.0061)	0.0090 (0.0093)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-35.25 [0.08]	-41.50 [0.01]	13.09 [0.51]	16.07 [0.33]
Dep. Var. Mean	0.089	0.156	0.103	0.185
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	260	260	260	260
Cell Clusters	3,897	3,897	3,897	3,897
Observations	77,940	77,940	77,940	77,940
	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Wet Season Phytomass	0.0004 (0.0013)	0.0027 (0.0030)		
Wet Season Phytomass $\times$ Transhumant Pastoral	-0.0061 (0.0050)	-0.0100 (0.0062)		
Dry Season Phytomass			-0.0001 (0.0014)	-0.0036 (0.0062)
Dry Season Phytomass $\times$ Transhumant Pastoral			-0.0017 (0.0047)	-0.0119 (0.0131)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-33.27 [0.22]	-30.96 [0.11]	-7.10 [0.72]	-28.16 [0.37]
Dep. Var. Mean	0.089	0.156	0.103	0.185
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	260	260	260	260
Cell Clusters	3,897	3,897	3,897	3,897
Observations	77,940	77,940	77,940	77,940

Table A15 Panel A: Summary of Seasonal Regressions, Non-Agricultural Cells Only, Rain

	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Rain	0.0057 (0.0117)	0.0082 (0.0204)	0.0039 (0.0126)	0.0246 (0.0192)
Rain $\times$ Transhumant Pastoral	-0.0197 (0.0189)	-0.2018 (0.1728)	-0.0137 (0.0156)	-0.1432 (0.1288)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain $\times$ Transhumant Pastoral p-value	-23.01 [0.30]	-102.26 [0.24]	-16.60 [0.38]	-78.22 [0.27]
Dep. Var. Mean	0.103	0.237	0.099	0.220
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	390	390	390	390
Cell Clusters	735	735	735	735
Observations	22,050	22,050	22,050	22,050
	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Wet Season Rain	0.0033 (0.0092)	-0.0230 (0.0270)		
Wet Season Rain $\times$ Transhumant Pastoral	-0.0045 (0.0152)	-0.0944 (0.0996)		
Dry Season Rain			0.0075 (0.0085)	0.0695 (0.0565)
Dry Season Rain $\times$ Transhumant Pastoral			-0.0039 (0.0124)	-0.0426 (0.0561)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-8.78 [0.77]	-80.91 [0.34]	-6.10 [0.75]	-29.59 [0.45]
Dep. Var. Mean	0.103	0.237	0.099	0.220
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	390	390	390	390
Cell Clusters	735	735	735	735
Observations	22,050	22,050	22,050	22,050

Table A15 Panel B: Summary of Seasonal Regressions, Non-Agricultural Cells Only, Phytomass

	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Annual Phytomass	0.0018 (0.0029)	-0.0032 (0.0101)	-0.0030 (0.0025)	-0.0051 (0.0071)
Annual Phytomass $\times$ Transhumant Pastoral	-0.0061 (0.0056)	-0.0565 (0.0480)	-0.0033 (0.0060)	-0.0414 (0.0385)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-15.57 [0.28]	-59.02 [0.24]	-9.14 [0.58]	-48.67 [0.28]
Dep. Var. Mean	0.130	0.317	0.119	0.282
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	260	260	260	260
Cell Clusters	735	735	735	735
Observations	14,700	14,700	14,700	14,700
	Wet Season UCDP Conflict		Dry Season UCDP Conflict	
	(1)	(2)	(3)	(4)
	Incidence Year Equiv.	No. Events Year Equiv.	Incidence Year Equiv.	No. Events Year Equiv.
<i>Nearest Neighboring Ethnic Group</i>				
Wet Season Phytomass	-0.0010 (0.0021)	-0.0173 (0.0154)		
Wet Season Phytomass $\times$ Transhumant Pastoral	-0.0033 (0.0038)	-0.0186 (0.0183)		
Dry Season Phytomass			0.0016 (0.0030)	0.0132 (0.0119)
Dry Season Phytomass $\times$ Transhumant Pastoral			-0.0092 (0.0079)	-0.0861 (0.0724)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Phytomass Shock as % of Dep. Var. Mean:				
Phytomass $\times$ Transhumant Pastoral p-value	-12.41 [0.39]	-28.38 [0.31]	-33.64 [0.24]	-133.23 [0.24]
Dep. Var. Mean	0.130	0.317	0.119	0.282
Cell FE	Yes	Yes	Yes	Yes
Country $\times$ Month FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	260	260	260	260
Cell Clusters	735	735	735	735
Observations	14,700	14,700	14,700	14,700

Table A17: Heterogeneity by the Presence of International Aid Projects: Sub-Categories

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral	-0.0115*** (0.0044)	-0.0117*** (0.0041)	-0.0103** (0.0050)	-0.0100** (0.0049)
Rain × Transhumant Pastoral × Irrigation Projects	0.0145 (0.0295)	-0.0066 (0.0284)	-0.0281 (0.0398)	-0.0261 (0.0395)
Rain × Transhumant Pastoral × Forestry Projects	0.0386* (0.0222)	0.0103 (0.0188)	0.0540 (0.0372)	0.0474 (0.0368)
Rain × Transhumant Pastoral × Conservation Projects	0.0079 (0.0273)	-0.0079 (0.0184)	-0.0188 (0.0355)	-0.0237 (0.0351)
Rain × Transhumant Pastoral × Land Projects	-0.0481 (0.0575)	-0.0071 (0.0532)	-0.0131 (0.0596)	-0.0146 (0.0594)
Rain × Transhumant Pastoral × Other Agriculture Projects	-0.0181* (0.0099)	-0.0067 (0.0088)	-0.0173 (0.0140)	-0.0146 (0.0138)
Rain × Transhumant Pastoral × Other Non-Agriculture Projects	0.0006 (0.0005)	0.0003 (0.0004)	0.0008 (0.0005)	0.0008 (0.0005)
Dep. Var. Mean	0.032	0.024	0.068	0.068
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	364	364	252	252
Cell Clusters	7,722	7,722	7,722	7,722
Observations	200,772	200,772	138,996	138,996

Table A18: Heterogeneous Treatment Effects, Controlling for Country FE and Year FE interacted with Main Interaction of Interest

	Panel A: Het by Int. Agricultural Aid			
	(1)	(2)	(3)	(4)
	UCDP I(Any)	UCDP I(State)	ACLED I(Any)	ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral × Total Agriculture Aid	-0.0161* (0.0083)	-0.0177** (0.0080)	-0.0031 (0.0114)	-0.0030 (0.0116)
Rain × Transhumant Pastoral × Total Non-Agriculture Aid	0.0011* (0.0006)	0.0011* (0.0006)	-0.0005 (0.0008)	-0.0005 (0.0008)
	Panel B: Het by Int. Aid Types			
	(1)	(2)	(3)	(4)
	UCDP I(Any)	UCDP I(State)	ACLED I(Any)	ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral × Irrigation Projects	0.0251 (0.0367)	-0.0069 (0.0346)	0.0342 (0.0524)	0.0378 (0.0520)
Rain × Transhumant Pastoral × Forestry Projects	0.0020 (0.0269)	-0.0132 (0.0255)	0.1487** (0.0656)	0.1422** (0.0651)
Rain × Transhumant Pastoral × Conservation Projects	0.0145 (0.0324)	-0.0045 (0.0231)	-0.0443 (0.0405)	-0.0491 (0.0398)
Rain × Transhumant Pastoral × Land Projects	-0.0890 (0.0606)	-0.0356 (0.0542)	-0.1500* (0.0786)	-0.1555** (0.0784)
Rain × Transhumant Pastoral × Other Agriculture Projects	-0.0157 (0.0129)	-0.0098 (0.0128)	-0.0159 (0.0188)	-0.0127 (0.0186)
Rain × Transhumant Pastoral × Other Non-Agriculture Projects	0.0009 (0.0007)	0.0009 (0.0007)	0.0003 (0.0008)	0.0003 (0.0008)
	Panel C: Het by Conservation, Country-Year Level			
	(1)	(2)	(3)	(4)
	UCDP I(Any)	UCDP I(State)	ACLED I(Any)	ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral × Share Protected Area in Country	-0.1398*** (0.0461)	-0.1508*** (0.0443)	-0.2368*** (0.0613)	-0.2340*** (0.0612)
	Panel D: Het by Conservation, Subnational-Year Level			
	(1)	(2)	(3)	(4)
	UCDP I(Any)	UCDP I(State)	ACLED I(Any)	ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral × Share Protected Area in Ethnicity e of Country c			0.0416** (0.0176)	0.0470*** (0.0166)
Rain × Transhumant Pastoral × Share Protected Area Outside of Ethnicity e in Country c			-0.1563*** (0.0522)	-0.1732*** (0.0499)
			-0.2714*** (0.0675)	-0.2678*** (0.0674)
	Panel E: Het by THP Political Power			
	(1)	(2)	(3)	(4)
	UCDP I(Any)	UCDP I(State)	ACLED I(Any)	ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain × Transhumant Pastoral × THP Power Share	0.0954** (0.0380)	0.0762** (0.0348)	0.2811*** (0.0755)	0.2810*** (0.0756)
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Rain × Transhumant Pastoralism × Country FE	Yes	Yes	Yes	Yes
Rain × Country FE	Yes	Yes	Yes	Yes
Rain × Transhumant Pastoralism × Year FE	Yes	Yes	Yes	Yes
Rain × Year FE	Yes	Yes	Yes	Yes
Transhumant Pastoralism × Year FE	Yes	Yes	Yes	Yes

Table A19: Heterogeneity by the Presence of Conservation Lands Using Subnational Variation

	Indicator for the presence of conflict			
	(1) UCDP I(Any)	(2) UCDP I(State)	(3) ACLED I(Any)	(4) ACLED I(Nonstate)
<i>Nearest Neighboring Ethnic Group</i>				
Rain	0.0036*** (0.0013)	0.0035*** (0.0012)	-0.0004 (0.0018)	-0.0005 (0.0018)
Rain × Transhumant Pastoral	-0.0104** (0.0048)	-0.0093** (0.0046)	-0.0018 (0.0054)	-0.0016 (0.0054)
Rain × Transhumant Pastoral × Share Protected Area in Ethnicity e of Country c	0.0574*** (0.0193)	0.0603*** (0.0181)	0.0546* (0.0278)	0.0530* (0.0279)
Rain × Transhumant Pastoral × Share Protected Area Outside of Ethnicity e in Country c	-0.0633* (0.0357)	-0.0816** (0.0351)	-0.1061*** (0.0367)	-0.1060*** (0.0368)
<i>Nearest Neighboring Ethnic Group: Additional Calculations</i>				
Effect of 1 Std. Dev. Rain Shock as % of Dep. Var. Mean:				
Rain × Transhumant Pastoral when Ethnicity Protected Area at 10th pctl	-35.6	-44.2	-2.6	-2.3
p-value	[0.03]	[0.04]	[0.73]	[0.76]
Rain × Transhumant Pastoral when Ethnicity Protected Area at 90th pctl	56.4	89.6	33.7	33.1
p-value	[0.12]	[0.05]	[0.08]	[0.09]
Rain × Transhumant Pastoral when Rest of Country Protected Area at 10th pctl	-45.2	-61.3	-9.3	-9.0
p-value	[0.00]	[0.00]	[0.18]	[0.20]
Rain × Transhumant Pastoral when Rest of Country Protected Area at 90th pctl	-109.5	-176.0	-54.0	-53.9
p-value	[0.00]	[0.00]	[0.00]	[0.00]
Dep. Var. Mean	0.035	0.025	0.085	0.084
Cell FE	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes
Climate-Zone-Year Clusters	420	420	322	322
Cell Clusters	7718	7718	7718	7718
Observations	231,540	231,540	177,514	177,514